AI Prompt Cheatsheet for Data Analysis

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How to use this cheatsheet

Every strong data prompt has five parts: **Role + Context + Data + Task + Format**. Copy a prompt below, replace the `[brackets]`, and paste your column list or a small sample.

Part	What to give the AI	Example
Role	Who the AI should act as	"Act as a senior data analyst in retail banking."
Context	Business goal and audience	"Management wants to know why Q3 sales dropped."
Data	Columns, data types, row count, sample rows	"Columns: Date, Branch, Product, Units, Revenue (12,000 rows)."
Task	One clear action verb	"Identify the top 3 drivers of the decline."
Format	Output shape and length	"Give a table, then 3 bullet insights under 20 words each."

Master template:

```
Act as a [role]. I am analysing [dataset] to [business goal] for [audience].
The data has these columns: [column: type, ...] with about [N] rows.
Task: [specific task].
Constraints: [tool/version, assumptions, what to avoid].
Output: [format, length, language]. Ask me questions first if anything is
unclear.
```

Tip: paste column headers plus 5–10 sample rows instead of the full file whenever the data is sensitive.

1. Data understanding & profiling

Goal	Prompt
First look	"Here are my column headers and 10 sample rows: [paste]. Describe what each column likely means, its data type, and what business questions this dataset can answer."
Data profile	"Profile this dataset: row count, missing values per column, unique counts, min/max, and obvious outliers. Present it as a table."
Data dictionary	"Create a data dictionary for these columns: [list]. Include name, description, type, example value and allowed values."
Grain check	"What is the grain of this table (one row = what)? Which columns form a unique key? How would I verify that?"
Question generator	"Suggest 10 business questions [department] could answer with this data, ranked by likely impact."
Quality red flags	"List data quality risks you see in this sample (duplicates, inconsistent formats, impossible values) and how to test each."

2. Data cleaning & preparation

Goal	Prompt
Cleaning plan	"Create a step-by-step cleaning plan for this data: [sample]. For each issue, state the fix and why."
Standardise text	"Column [Name] has inconsistent values like [examples]. Give rules to standardise them into a clean list."
Missing values	"Column [X] has [N]% missing. Recommend whether to drop, fill, or flag, with pros and cons for [business purpose]."

Goal	Prompt
Duplicates	"How do I find and remove duplicates where [columns] match but [column] differs? Give steps in [Excel / Power Query / SQL]."
Power Query steps	"Write Power Query (M) steps to: remove top [N] rows, promote headers, unpivot [columns], change types, and trim text."
Combine files	"I have monthly files in one folder with the same layout. Explain how to combine them in Power Query and add the file name as a Month column."
Split & parse	"Split column [X] containing values like [example] into [parts]. Give the formula or M code."
Date fixes	"My dates appear as [examples] in mixed formats. How do I convert them to proper dates in [tool]?"

3. Exploratory analysis & statistics

Goal	Prompt
EDA plan	"Design an exploratory analysis plan for [dataset] to answer [question]. List the cuts, metrics and charts in order."
Trend	"Analyse the trend of [metric] by [month/quarter]. Describe growth rate, seasonality and any unusual periods."
Segmentation	"Compare [metric] across [segment column]. Which segments over- or under-perform the average, and by how much?"
Root cause	"[Metric] dropped [X]% from [period A] to [period B]. Break down the change by [dimensions] to find the main drivers."
Correlation	"Which variables are most related to [target]? Explain correlation vs causation for each finding."

Goal	Prompt
Right test	"I want to test whether [A] differs from [B] using [data type]. Which statistical test should I use, what are its assumptions, and how do I read the result?"
Explain simply	"Explain this result to a non-technical manager in 3 sentences: [paste output]."
Outliers	"Identify outliers in [column] using IQR and z-score. Should they be removed, capped or investigated?"

4. Excel formulas & PivotTables

Goal	Prompt
Write a formula	"Write an Excel formula that [task]. My data is in [range], headers in row 1, using Excel [365/2019]. Explain each part."
Lookup	"Use XLOOKUP to return [column] from [Sheet2] matching [key], and show '[Not found]' when there is no match."
Fix an error	"This formula returns [#N/A / #VALUE!]: [formula]. Why, and how do I fix it?"
Dynamic arrays	"Using FILTER, SORT and UNIQUE, build a list of [items] where [condition], sorted by [column] descending."
PivotTable design	"I need a PivotTable showing [metric] by [rows] and [columns], filtered by [field]. Which fields go where, and which calculated field do I need?"
Power Pivot measure	"Write a DAX measure in Power Pivot for [metric], e.g. year-on-year growth, using tables [Fact] and [Date]."
Automate	"Write a VBA macro / Office Script that [repetitive task]. Add comments so I can maintain it."
Dashboard layout	"Suggest an Excel dashboard layout for [audience] with KPIs, slicers and 3–4 charts for [data]."

5. Power BI, DAX & data modelling

Goal	Prompt
Star schema	"My tables are [list with columns]. Design a star schema: fact vs dimension tables, keys, relationships and cardinality."
Date table	"Write DAX to create a Date table from [start] to [end] with Year, Quarter, Month name, Month number and Financial Year starting [month]."

Goal	Prompt
DAX measure	"Write a DAX measure for [metric] using [table/columns]. Explain the filter context step by step."
Time intelligence	"Write measures for MTD, QTD, YTD, previous year and YoY % for [measure], using 'Date'[Date]."
Debug DAX	"This measure returns [wrong result]: [DAX]. Expected [result]. Find the error and correct it."
CALCULATE help	"Explain what CALCULATE, FILTER, ALL and REMOVEFILTERS do in this measure: [DAX]."
Ranking / Top N	"Create a measure that ranks [dimension] by [measure] and a Top N visual filter driven by a slicer."
Performance	"My report is slow. Model: [details]. Suggest optimisation steps for the model, DAX and visuals."
RLS	"Set up row-level security so each [manager] only sees [their region]. Give the DAX role filter."

6. SQL & Python

Goal	Prompt
Write SQL	"Using [SQL Server / MySQL / PostgreSQL], write a query that returns [result] from tables [schema]. Comment each step."
Joins	"Tables [A] and [B] share [key]. Which join keeps [requirement]? Write the query and show expected row behaviour."
Window functions	"Write SQL with window functions for running total, rank and month-over-month change of [metric] by [group]."
Optimise	"This query is slow on [N] rows: [SQL]. Suggest indexes and rewrites."

Goal	Prompt
Python EDA	"Write pandas code to load [file], clean [issues], and produce summary stats and charts for [columns]."
Explain code	"Explain this code line by line for a beginner, then list possible bugs: [code]."
Convert	"Convert this Excel logic / SQL into pandas (or DAX): [paste]."
Error fix	"I get this error: [traceback]. Here is my code: [code]. What causes it and what is the fix?"

7. Visualisation & dashboard design

Goal	Prompt
Choose a chart	"I want to show [message] using [fields]. Which chart type fits best, and which should I avoid?"
KPI selection	"Suggest 5–7 KPIs for a [department] dashboard aimed at [audience], with formula and target logic for each."
Layout wireframe	"Sketch a one-page dashboard layout for [goal]: KPI cards, main chart, breakdowns, slicers, in reading order."
Critique	"Critique this dashboard description/screenshot for clarity, clutter, colour use and accessibility. Give 5 fixes."
Colour & theme	"Create a Power BI JSON theme using [brand colours], with colours for good, warning and bad."
Titles that tell	"Rewrite these chart titles as insight headlines: [titles + key numbers]."
Looker Studio	"How do I build [chart/calculated field/blended data] in Looker Studio from [source]?"

8. Insights, storytelling & reporting

Goal	Prompt
Key insights	"From these results [paste], give the top 5 insights. For each: what happened, why it matters, and a recommended action."
Executive summary	"Write a 150-word executive summary for [audience] covering result, cause, impact in RM, and next step."
So what?	"For each finding below, answer 'So what?' and 'Now what?' in one line each: [findings]."
Story structure	"Structure a data story for [meeting] using Situation, Complication, Resolution, with the chart for each part."
Slide outline	"Turn this analysis into a 6-slide outline: title, key message headline, supporting chart and speaker notes."
Email update	"Write a brief email to [stakeholder] summarising [results] with 3 bullets and one ask."
Challenge me	"Act as a sceptical CFO. List 5 tough questions about this analysis and how I should answer them."
Bahasa version	"Terjemahkan ringkasan ini ke Bahasa Melayu formal untuk laporan pengurusan: [text]."

9. Forecasting & machine learning

Goal	Prompt
Pick a method	"I have [N] months of [metric] with [seasonality/trend]. Recommend a forecasting method (e.g. moving average, ETS, ARIMA, Prophet) and explain why."
Excel forecast	"How do I use FORECAST.ETS or Forecast Sheet in Excel to predict [metric] for the next [N] periods, with confidence intervals?"

Goal	Prompt
Model framing	"Frame this business problem as an ML task: [problem]. Target variable, features, model type and success metric?"
Feature ideas	"Suggest 15 features I could engineer from [columns] to predict [target]."
Evaluate model	"Explain these results to a manager: accuracy [x], precision [x], recall [x], AUC [x]. Is the model good enough for [use case]?"
Text analytics	"Outline a topic modelling (LDA) workflow for [N] customer comments: cleaning, choosing topic count, and labelling topics."
Explainability	"How can I explain which features drive this model's predictions to non-technical stakeholders?"

10. Validation, follow-ups & data privacy

AI answers about data must be checked before they reach a report. Use these follow-up phrases to sharpen or verify any output.

Follow-up phrase	When to use it
"Show your calculation step by step."	Any number you will report
"What assumptions did you make?"	Before trusting a conclusion
"Give me a way to verify this in [Excel/Power BI]."	Checking AI figures against the source
"What could make this conclusion wrong?"	Stress-testing insights
"Make it shorter / for a non-technical audience."	Adjusting tone and length
"Give 3 alternative approaches and compare them."	Choosing a method
"Ask me 5 clarifying questions first."	Vague or complex tasks
"Format as a table I can paste into Excel."	Reusing output

Data privacy checklist

- [ ] Remove or mask names, IC numbers, phone numbers, emails and account numbers before pasting
- [ ] Share headers and sample rows instead of full datasets
- [ ] Use your organisation's approved AI tool (e.g. Microsoft Copilot with enterprise data protection) for confidential data
- [ ] Re-check every AI-generated number against the source before publishing
- [ ] Keep the final judgement with a human: AI drafts, the analyst decides